

Supplementary Material

Degree-Independent Control Impact in the Human Structural Connectome:
Graph-Geometric Associations and Degree-Preserving Null Arbitration (Paper 2, v2.0.1)

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About this document. Tables S0–S10 are auto-generated from the frozen repository artifacts by `13_MANUSCRIPT/make_supplementary.py` and stored as `SUPPLEMENTARY_TABLES.md` (machine-readable, regenerate with `python run_all.py`). This PDF re-typesets the decision-relevant subset; numbers are identical by construction.

1 S0. Protocol amendments and timing

Table 1: All amendments were pre-unblinding implementation repairs; no frozen gate, threshold, seed, or hypothesis was altered.

Am.	Problem	Discovery point	Resolution
A1	$\log_{10}(\text{CIS} + 10^{-6})$ estimator invalid: CIS is signed; log of negative CIS $\rightarrow$ NaN	first Stage-3 execution (QC figure failure)	estimator re-specified pre-unblinding
A2	(1) asinh retransformation biased (Jensen; $\rho(R, \text{deg}) = -0.063$); (2) cross-check unit mismatch (all-node vs pair-only)	pre-unblinding gate check	final M1 = raw-CIS spline df=4; same-pairs cross-check
A3	redundancy feature used boolean adjacency (OR-semantics): redundancy $\equiv$ clustering on all rows	pre-analysis integrity sweep (Stages 7+ not run)	int32 cast; caches re-generated; unit assertion added
FREEZE	study completion	2026-09-30	study tree declared read-only

2 S2. Univariate associations (Stage 8, all 12 features)

3 S6. Degree-preserving null arbitration (Stage 15)

Verdict: Outcome C. No observed association exceeds its null p95 band; the mechanism claim is null-qualified.

4 S6b. Null-estimator (proxy) equivalence validation

Median $|\Delta\rho| = 0.050$ over 12×9 subject–feature pairs vs null bands of 0.10–0.73. Point-level residuals differ in the tails (median $\text{Spearman}(R_{\text{spline}}, R_{20}) = 0.865$); both are degree-orthogonal ($|\rho(R, \text{deg})| \leq 0.041$).

Table 2: Within-subject median Spearman $\rho(R, \cdot)$; q = BH-FDR. All features $q < 3.4 \times 10^{-28}$. Values quoted verbatim from 05_MECHANISM_ANALYSIS/UNIVARIATE_RESULTS.json.

Feature	median ρ	IQR	q
bridge	+0.4985	0.1007	$< 10^{-300}$
closeness_d	+0.2489	0.0770	$< 10^{-300}$
redundancy	-0.2179	0.0835	$< 10^{-300}$
participation	-0.1676	0.1310	5.5×10^{-256}
clustering	-0.1658	0.0888	$< 10^{-300}$
betweenness	+0.1333	0.0759	$< 10^{-300}$
eigenvector	+0.0834	0.0745	6.5×10^{-199}
strength	-0.0689	0.0722	2.3×10^{-173}
kcore	+0.0556	0.1352	6.1×10^{-44}
pagerank	-0.0379	0.0555	1.1×10^{-110}
degree	-0.0255	0.0534	8.9×10^{-58}
within_module_z	-0.0186	0.0647	3.4×10^{-28}

Table 3: 12 subjects $\times$ 100 nulls, seed 20270927; exact degree preservation verified per null; full CIS + feature recomputation per null. Deep-verified from the 1,200 raw records by independent gate V09 (seeds unique; degree preservation exact; stored R_{null} re-derives bit-exactly via the 20-bin proxy; pooled statistics reproduce to 10^{-9}).

Feature	Obs. med ρ	Null med ρ	Null IQR	Null $ \rho $ p95	Verdict
bridge	0.4985	-0.5957	0.0502	0.6538	inside band
clustering	0.1658	+0.6846	0.0443	0.7314	inside band
redundancy	0.2179	-0.6228	0.0469	0.6807	inside band
participation	0.1676	+0.3909	0.0695	0.4609	inside band
betweenness	0.1333	+0.1675	0.0342	0.2102	inside band
within-module z	0.0186	+0.0942	0.0448	0.1499	inside band
eigenvector	0.0834	+0.0926	0.0317	0.1304	inside band
k-core	0.0556	+0.0627	0.0323	0.1003	inside band
PageRank	0.0379	+0.0618	0.0304	0.0961	inside band

5 Remaining tables (machine-readable)

The following are provided in SUPPLEMENTARY_TABLES.md (generated, artifact-backed): **S1** feature definitions; **S3** feature correlation matrix (prune rule $|\rho| \geq 0.9$); **S4** nested models; **S5** subject-aware CR1 regression; **S7** robustness matrix (R1/R2/R5); **S8** full biological enrichment (all categories, both sets); **S9/S9b** chokepoint candidates and threshold sensitivity; **S10** human-vs-fly comparison (fly negative preserved verbatim).

Table 4: 20-bin quantile-median proxy vs frozen spline on observed data (12 null subjects; Stage-8 features). The arbitrated statistic $\rho(R, \cdot)$ is stable across residualizations; the proxy yields slightly *stronger* associations, so it cannot manufacture Outcome C.

Feature	med ρ (spline)	med ρ (20-bin)	med $ \Delta\rho $	p95 $ \Delta\rho $
bridge	0.4661	0.5206	0.0436	0.1258
betweenness	0.1028	0.1425	0.0504	0.1333
clustering	-0.1469	-0.1686	0.0479	0.1319
eigenvector	0.1078	0.1331	0.0480	0.1341
kcore	0.0361	0.0750	0.0647	0.1354
pagerank	-0.0460	-0.0074	0.0466	0.1450
participation	-0.1173	-0.0697	0.0639	0.1041
redundancy	-0.1826	-0.2011	0.0467	0.1158
within-module z	-0.0211	+0.0114	0.0482	0.1417